CS & IT ENGINEERING

'C' Programming

Data Types & I/O Functions

Lecture No.- 02

By- Satya sir

Recap of Previous Lecture

- Operator Precedence & Associativity
- Ternary operator
- Data Type definition
- Classification of Data Types
    - Primary datatypes

Topics to be Covered

- Primary datatypes
  - int, float, char, void
- I/O Functions

Primary Data Types : int, char, float, void

Void type : No specific type

Integer datatype : int data members;

Ex:

```
int a, b, c ;        -> Variables [data whose value may change]
int *P, *q;          -> Pointers
int fun( );          } return type
(int *) fun( ) ;     }
int x[3], y[4];      -> Arrays
```

```
int a=~~7~~ 5 3, b=~~3~~ 2;
a=a+b; => a=5
b=a-b; => b= 2
a=a-b => a=3
```

Integer datatype

Nature: A data whose value is a whole number [A Number without fractional Part]

Ex: 4, 7, 13, 147, 2983 - - -

Memory size : It is Compiler - dependant in 'C' Language

16bit Processor / 16-bit Compiler ⇒ 1 integer = 16 bits (or) 2 Bytes

32-bit Processor / 32-bit Compiler ⇒ " = 32 bits (or) 4 Bytes

64-bit Processor / 64-bit Compiler ⇒ " = 64 bits (or) 8 Bytes

Range of values : if integer size is 'i' bits

Type Extensions : Signed, Unsigned, short, long

⇒ Short int a; (or) short a;

size = $i/2$ bits

⇒ Signed int (or) int a; ($\pm$ve values)

Ex: Signed int GATE_marks;

⇒ Unsigned int x; (+ve only)

Ex: Unsigned int age;

⇒ long int (or) long a

Memory Size = $2*i$ bits

x bits $\Rightarrow$ Total 2^x values Possible.
Range(+ve) 0 to $(2^x - 1)$

Range($\pm$ve): $\left(-\frac{2^x}{2} \ldots\ldots -1\right)\left(0 \ldots +\left(\frac{2^x}{2}-1\right)\right)$

4 Types of Integers

[0---9] ① Decimal Integer → Any whole Number is by default, decimal Integer int a=72;

[0---7] ② Octal Integer ⇒ Value with Prefix '0' (zero) ⇒ int a=072;

[0---F] ③ Hexa Decimal Integer ⇒ Value with Prefix '0x' ⇒ int a=0x72;

[0-1] ④ Binary Integer ⇒ value with 1's and 0's, with Prefix '0b' ⇒ int a=0b01001000;

Range

Memory size is 'x' bits → unsigned form 0 to $(2^x - 1)$
→ Signed form $-2^{(x-1)}$ to $\left[2^{(x-1)} - 1\right]$

Ex: x=16bits ⇒ unsigned int a; Range: 0 to $(2^{16}-1)$ ⇒ 0 to 65535
⇒ signed int a; -2^{15} to $(+2^{15}-1)$ ⇒ -32768 to +32767

1 bit

0 } Total
1 } 2 values

2 bits

0	0	⇒ 0
0	1	⇒ 1
1	0	⇒ 2
1	1	⇒ 3

4 val

3 bits

000	001	010	011	100	101	110	111
0	1	2	3	4	5	6	7

8 values

0-48
1-49
2-50
3-51
4-52
5-53
6-54
7-55
8-56
9-57

Character DataType

Nature : dataitem, whose value can be either an alphabet (or) Symbol (or) Single digit number.

- characters are represented using single quotes (' ')

  Ex: 'x', '7', '@', '#', 'p', 'A' - - -

- char is keyword. Syntax: char dataitem;

  char x='7', y='@', ch='p';

- In 'C' Language, Every character is Processed | Identified by Compiler with it's Unique ASCII Value.

  Ex: 'A' = 65, 'a' = 97 '0' = 48

- Total 256 ASCII Values ⇒ So, $\boxed{\text{Character Memory Size == 8 bits (or) 1 Byte, in 'C' Language.}}$

- Type Extensions for character : signed, unsigned

signed char ch; (or) char ch; Range : $-2^{(8-1)}$ to $+\left[2^{(8-1)}-1\right]$ $= \underline{\underline{-128 \text{ to } +127}}$

unsigned char ch; Range : 0 to $[2^8-1]$ $\Rightarrow$ 0 to 255

Float datatype

Nature : A Number with fractional Part (or) Precision (or) • symbol.

Ex: 3.4, 1.235, 0.01234, 27.

Type Extension : long

long float ≅ double

Memory Size : float = 4 Bytes (or) 32 bits

long float (or) double ⇒ 8 Bytes (or) 64 bits

long long float (or) long double ⇒ 10 Bytes (or) 80 bits

No. of bits Precision	Range
6 bits	$3.4 * 10^{-38}$ to $3.4 * 10^{+38}$
15 bits	$1.7 * 10^{-308}$ to $1.7 * 10^{+308}$
17 bits	$1.7 * 10^{-4932}$ to $1.7 * 10^{+4932}$

I/O Functions in 'C'

- The statements, that cause input to be given by User and output displayed for User, are said to be i/o statements.

- 2 Types
  1. Console i/o functions : i/p ⟵ Keyboard, o/p ⟶ monitor (or) display device
  2. Disk (or) File i/o functions : i/o ⟷ File in Memory : fprintf(), fscanf(), fgetw(), fputw(), fgets(), fputs() - - .

Console i/o functions

a) formatted i/o functions : scanf(), Printf()

b) Unformatted i/o functions : getch(), getchar(), gets(), Putchar(), Puts()

To be Contd . . .

2 mins Summary

- Data Types in 'C'
  - Primary datatypes
    - void
    - char
    - int
    - float, double
- I/O functions introduction.

THANK - YOU